

Jiayi Zhang

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Education

University of Nottingham Ningbo China (UNNC)

BSc (HONS) MATHEMATICS WITH APPLIED MATHEMATICS

- GPA - 3.94/4.00

Ningbo China

Sep 2024 - Jun 2028 (Expected)

Research Interests

My research interests include **neuro-symbolic robotic manipulation**, **geometry-aware decision-making**, **long-horizon planning**, and **foundation model-based robot learning**. I am interested in combining **neuro-symbolic planning**, **Task and Motion Planning (TAMP)**, and **Vision-Language-Action (VLA) models** to build robots that can reason, plan, and act under **uncertainty** in complex **human-centered tasks**, such as **table-top Mahjong playing**.

Experience

Tsinghua University (THU)

RESEARCH ASSISTANT, ADVISED BY [PROF. MENGDI XU](#)

- Research Topics:** Manipulation, Neuro-Symbolic planning, VLA models

Beijing, China

July 2026- Now

Shanghai Jiao Tong University (SJTU)

RESEARCH ASSISTANT AT [IWIN-FINS LAB](#), ADVISED BY [PROF. JIANPING HE](#)

- Research Topics:** optimal control, generative model, grasping
 - Robot arm:** Project leader
 - D.I.V.E FORUM:** Organizer
 - Underwater 3D reconstruction:** Project leader
 - WBCD 2nd dual arm robot:** member

Shanghai, China

June 2025 - June 2026

The University of Hong Kong (HKU)

RESEARCH INTERN AT [IoT LAB](#), ADVISED BY [PROF. EDITH C.H. NGAI](#)

- Research Topics:** federated learning, fairness, spatio-temporal graphs.

Hong Kong SAR, China

March 2025 – August 2025

Publications

PREPRINTS

[P1] **F²STNet: Fair and Federated Spectral-Temporal Modeling for Graph Forecasting**

Jiayi Zhang*, Jinfeng Xu*, Hewei Wang*, Siyuan Cen, Haidong Huang, Yiyao Zhan, Zheyu Chen, Jinjiang You, Ai Jian, Edith C. H. Ngai
Under review, 2025

CONFERENCE PROCEEDINGS

[C4] **Multi-modal Dynamic Proxy Learning for Personalized Multiple Clustering**

Jinfeng Xu, Zheyu Chen, Shuo Yang, Jinze Li, Ziyue Peng, Zewei Liu, Hewei Wang, [Jiayi Zhang](#), Edith C. H. Ngai
AAAI, 2026

[C3] **Multi-Cali Anything: Dense Feature Multi-Frame Structure-from-Motion for Large-Scale Camera Array Calibration**

Jinjiang You*, Hewei Wang*, Yijie Li, Mingxiao Huo, LVT Ha, Mingyuan Ma, Jinfeng Xu, [Jiayi Zhang](#), Puzhen Wu, Shubham Garg, Wei Pu
IROS, 2025 ([Oral](#)) [[Paper](#)]

[C2] **Spec-LLAVA: Accelerating Vision-Language Models with Dynamic Tree-Based Speculative Decoding**

Mingxiao Huo*, [Jiayi Zhang](#)*, Hewei Wang*, Jinfeng Xu, Zheyu Chen, Huilin Tai, Ian Yijun Chen
ICML Workshop, 2025 [[Paper](#)]

[C1] **Segregation and Context Aggregation Network for Real-Time Cloud Segmentation**

Yijie Li*, Hewei Wang*, [Jiayi Zhang](#), Jinjiang You, Jinfeng Xu, Puzhen Wu, Yunzhong Xiao, Soumyabrata Dev
ICLR Workshop, 2025 [[Paper](#)]

JOURNAL ARTICLES

[J2] **TrafficKAN-GCN: Graph Convolutional-based Kolmogorov-Arnold Network for Traffic Flow Optimization**

[Jiayi Zhang](#)*, Yiming Zhang*, Yuan Zheng, Yuchen Wang, Jinjiang You, Yuchen Xu, Wenxing Jiang, Soumyabrata Dev
arXiv preprint, 2025 [[Paper](#)]

Honors & Awards

2026	Outstanding student (Top 1% student) , UNNC Award for Outstanding Students (UG)	Ningbo China
2025	Provost's Scholarship (Top 5% student) , University Academic Excellence Scholarship	Ningbo China
2025	Third Prize & Best Researcher Award , 9th Golden Interns Award	Ningbo China
2025	Honorable Mention , Interdisciplinary Contest in Modeling 2025	Online

Academic Services

Reviewer NeurIPS, AAAI, AISTATS 2025–Present

Leadership/Mentorship/Social

Semester Peer Mentor

UNIVERSITY OF NOTTINGHAM NINGBO CHINA (UNNC)

- Provided academic guidance and practical resources for freshmen.

Ningbo China
Sep. 2025–Jun. 2026

Skills

Programming Python, C++, $\text{\LaTeX}$, MATLAB, HTML, CSS, R

Frameworks PyTorch, Tensorflow, NumPy, Flask, MySQL, Git, Anaconda, OpenCV, ROS

Models CNN, ResNet, Faster R-CNN, YOLO, LSTM, Transformer, BERT, ViT, GANs, VAE, Stable Diffusion

Platforms Unitree Z1 Pro, Franka Research 3

Additional

Language Chinese (native), English (working proficiency)

Interests Running, hiking, mahjong, chessing, pickleball, roguelike and strategy games